



ITSimplera Institute

Red Team Internship — Week 01 Assignment

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Submission Date
03 July 2026

What is Cybersecurity?

Definition

- The practice of protecting computers, networks, applications, and digital information from cyber threats.
- It ensures systems remain secure by preventing unauthorised access, maintaining data accuracy, and keeping services available to authorised users.
- Cybersecurity combines people, processes, and technology to identify risks, defend against attacks, and minimize the impact of security incidents on individuals and organizations.

Why It Matters

- ☐ Protects sensitive data from theft and exposure
- ☐ Prevents cyberattacks before they cause damage
- ☐ Reduces financial losses from breaches
- ☐ Maintains business continuity under threat
- ☐ Builds customer trust and brand reputation
- ☐ Supports legal and regulatory compliance

The CIA Triad

Three foundational principles that guide every cybersecurity decision — from access control to disaster recovery.



Confidentiality

Prevents unauthorised access to sensitive information through encryption and access controls.



Integrity

Ensures information remains accurate, trustworthy, and unaltered throughout its lifecycle.



Availability

Guarantees systems and data remain accessible to authorised users whenever required.

Real-World Cyberattacks

Five landmark incidents that redefined enterprise security strategy worldwide.

WannaCry — 2017

Target: NHS, United Kingdom

Type: Ransomware

Exploited the EternalBlue vulnerability in Windows, encrypting files and demanding Bitcoin ransom.

Impact: 200,000+ systems infected across 150+ countries

CIA Violated: Availability

Equifax Breach — 2017

Target: Equifax, United States

Type: Data Breach

Attackers exploited an unpatched Apache Struts vulnerability to access customer records.

Impact: 147 million individuals' data exposed

CIA Violated: Confidentiality, Integrity

SolarWinds — 2020

Target: U.S. Government agencies

Type: Supply Chain Attack

Malicious code inserted into legitimate software updates, compromising thousands of organisations.

Impact: Thousands of organisations compromised

CIA Violated: Confidentiality, Integrity

Colonial Pipeline — 2021

Target: Colonial Pipeline, USA

Type: Ransomware

DarkSide ransomware forced shutdown of pipeline operations, causing fuel shortages.

Impact: Major fuel supply disruption

CIA Violated: Availability

MOVEit Transfer — 2023

Attack Profile

Target: Organisations using MOVEit Transfer worldwide

Type: Zero-Day Exploit (SQL Injection)

CloP ransomware exploited a SQL injection vulnerability to exfiltrate data.

Impact & Response

- Millions of records compromised across thousands of organisations
- Emergency patches deployed by Progress Software
- Incident response teams activated globally
- Continuous monitoring and threat hunting initiated

❏ **CIA Violated:** Confidentiality & Integrity — sensitive data exposed and altered.

CIA Triad in Practice

Each principle maps to specific controls and connects directly to real-world attack outcomes.

Confidentiality

Controls: Encryption, MFA, Access Control, Strong Passwords

Related Attacks: Equifax, SolarWinds, MOVEit

Integrity

Controls: Hashing, Digital Signatures, Checksums, File Integrity Monitoring

Related Attacks: Equifax, SolarWinds, MOVEit

Availability

Controls: Backups, Disaster Recovery, Load Balancing, Redundant Systems

Related Attacks: WannaCry, Colonial Pipeline

Cybersecurity Tools

Three essential tools every red team analyst must master.

Wireshark

Purpose: Network packet analysis

Function: Captures and analyses live network traffic in real time.

Use Case: Malware investigation, troubleshooting, incident response.

Nmap

Purpose: Network discovery

Function: Identifies hosts, services, operating systems, and open ports.

Use Case: Vulnerability assessment and penetration testing.

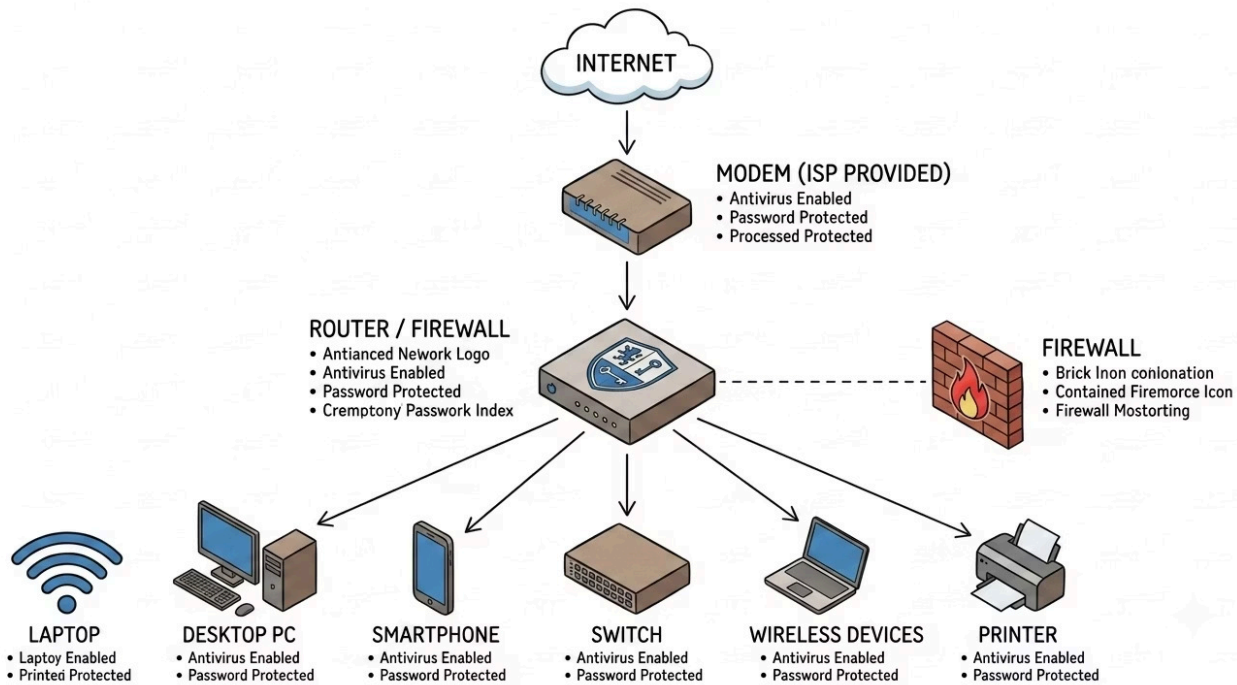
Burp Suite

Purpose: Web application security testing

Function: Intercepts and analyses HTTP/HTTPS requests.

Use Case: Detecting SQL Injection, XSS, and authentication flaws.

Secure Home Network



Security Checklist

WPA3 Wi-Fi
Encryption

Strong Router
Password

Firewall
Enabled

Antivirus
Installed

Automatic
Updates

Guest Wi-Fi
Network

Analyst Responsibilities

A cybersecurity analyst's daily operations span monitoring, investigation, and continuous improvement.

Monitor SIEM Alerts

Investigate Incidents

Vulnerability Assessments

Analyse Firewall & IDS/IPS Logs

Detect Malware & Phishing

Apply Security Patches

Conduct Risk Assessments

Maintain Documentation

Monitor Network Traffic

Research Emerging Threats

Recommend Improvements

Collaborate with IT Teams

Challenges & Conclusion

1

Ransomware attacks

2

Phishing campaigns

3

Zero-day vulnerabilities

4

Insider threats

5

Cloud security risks

Conclusion

Cybersecurity is essential for protecting digital assets and ensuring the confidentiality, integrity, and availability of information.

Understanding real-world attacks, leveraging the right tools, and embracing analyst responsibilities together provide a strong foundation for defending modern organisations against an evolving threat landscape.

References: (1) CISA.gov — WannaCry & Colonial Pipeline Advisories. (2) US-CERT — SolarWinds Supply Chain Compromise. (3) FTC — Equifax Data Breach Settlement. (4) Progress Software — MOVEit Transfer Security Advisories.